CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 10: Arithmetic Progression

Every formula and result of the chapter on one sheet

1	General A.P.	$a,\ a+d,\ a+2d,\ \ldots$
2	Common difference	$d = t_n - t_{n-1}$
3	General term	$t_n = a + (n-1)d$
4	nth term from the end	$l - (n-1)d$
5	Number of terms	$n = \frac{l-a}{d} + 1$
6	Sum of n terms	$S_n = \frac{n}{2}\left[2a + (n-1)d\right]$
7	Sum with last term known	$S_n = \frac{n}{2}(a+l)$
8	Term from the sums	$t_n = S_n - S_{n-1}$
9	Arithmetic mean of a and b	$\frac{a+b}{2}$
10	Condition for A.P.	$2b = a + c$
11	Three terms in A.P.	$a-d,\ a,\ a+d$
12	Sum of the first n natural numbers	$\frac{n(n+1)}{2}$